

SETS - HOMEWORK

1. Enumerate the elements of the following sets:

- (a) $A = \{2n + 1 : n \text{ is a positive integer}\}$
- (b) $B = \{n^2 : n \text{ is an odd integer}\}$
- (c) $C = \{x^2 + 1 : x \text{ is an integer and } 3 \leq x < 6\}$
- (d) $D = \{x : x \in \mathbb{Z}, x^2 < 20\}$
- (e) $E = \{x : x \in \mathbb{R}, x^2 - 2x - 15 = 0\}$

2. Express each of the following sets in predicate form:

- (a) $A = \{5, 10, 15, 20, \dots\}$
- (b) $B = \{1, 5, 9, 13, 17, \dots\}$
- (c) $C = \{\frac{1}{3}, \frac{1}{8}, \frac{1}{15}, \dots\}$

3. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$,

$$A = \{2, 4, 6, 8, 10, 12\},$$

$$B = \{3, 6, 9, 12\} \text{ and}$$

$$C = \{1, 2, 5, 6, 9, 10\}.$$

Find (a) $A \cap C$, (b) $B \cup C$, (c) C^c , (d) $B - C$, (e) $B \triangle C$ and (f) $B \cap C$ and draw the corresponding Venn Diagram.

4. Let $U = \{n : n \text{ is an integer and } 1 \leq n \leq 11\}$,

$$A = \{2n + 1 : n \text{ is an integer and } 1 \leq n \leq 5\},$$

$$B = \{n : n \text{ is an integer and } 1 \leq n \leq 7\} \text{ and}$$

$$C = \{n : n \text{ is an integer and } 4 \leq n \leq 10\}.$$

Find (a) $A \cap B$, (b) $A \cup C$, (c) B^c , (d) $A - C$, (e) $A \triangle C$, (f) $(A \cup C) \cap B$, (g) $(B \cap C^c) \cup A^c$, (h) $(A^c \cup B)^c \cup C^c$, (i) $(A \cup B)^c \triangle (B \cup C)$ and draw the corresponding Venn Diagram.

5. Use the laws of the algebra of sets to prove that for any sets A , B and C

- (a) $(A \cap B^c)^c \cup B = A^c \cup B$
- (b) $(A^c \cap (B \cup C)^c)^c = A \cup B \cup C$
- (c) $(A - B) - C = A - (B \cup C)$

6. Let the universal set $U = \{p, q, r, s, t, u, v, w\}$. Let $A = \{p, q, r, s\}$, $B = \{r, t, v\}$ and $C = \{p, s, t, u\}$. Find the elements of the following sets: (a) $B \cap C$, (b) $A \cup C$, (c) C^c , (d) $A \cap B \cap C$, (e) $(A \cup B) \cap (A \cap C)$, (f) $(A \cup B)^c$, (g) $B - C$ (h) $B \triangle C$ and (i) $(A \triangle C) \cap B^c$

7. Consider the following subsets of $\mathbb{Z}$:

$$A = \{3n : n \in \mathbb{Z}\}$$

$$B = \{2n : n \in \mathbb{Z}\}$$

$$C = \{n : n \in \mathbb{Z} \text{ and } n^2 \leq 100\}$$

Using set operations, express each of the following in terms of A , B and C :

- (a) the set of all odd integers
 - (b) $\{-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10\}$
 - (c) $\{6n : n \in \mathbb{Z}\}$
 - (d) $\{-9, -7, -5, -3, -1, 1, 3, 5, 7, 9\}$
 - (e) $\{-6, 0, 6\}$
 - (f) $\{-9, -3, 3, 9\}$
 - (g) $\{-7, -5, -1, 1, 5, 7\}$
8. Each of the 40 first-year students can study a number of optional units. If 20 chose to study mathematics, 23 chose to study statistics and 5 took neither mathematics nor statistics, how many students studied both mathematics and statistics?
9. Each of the 32 first-year students can study a number of optional units. Suppose 7 chose to study accounting, 17 chose to study business and 14 chose to study computer science. Suppose that 2 students took all three options. Suppose that 3 students studied both accounting and business, 3 studied both accounting and computer science and 7 studied both business and computer science.
- (a) How many students are taking at least one of the three options?
 - (b) How many students are taking none of the three options?
 - (c) How many students only took business?
 - (d) How many students are taking accounting but not business?
 - (e) How many students are taking accounting or computer science, but not both.
10. Let $A = \{a, b, c\}$ and $B = \{x, y, z, w\}$. Find the Cartesian products $A \times B$, $B \times A$ and $A \times A$. What is the cardinality of the set $B \times B$?
11. Let $S = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 4, 5\}$ and $B = \{3, 5\}$. Find the characteristic vectors of A and B , and determine the characteristic vectors of $A \cap B$, $A \cup B^c$ and $A \triangle B$.